

DEEP REINFORCEMENT LEARNING — QUICK REVISION SHEET

Main Concepts · Formulas · Numerical Steps

Foundations

RL: learn actions to maximise cumulative reward by trial-and-error; *delayed* evaluative feedback.

Reward hypothesis: any goal = maximising expected cumulative scalar reward.

Elements: policy, reward, value, model.

Explore vs exploit: gather info vs use best known.

Multi-Armed Bandits

Sample average: $Q_{n+1} = Q_n + \frac{1}{n}(R_n - Q_n)$; constant α for non-stationary.

ϵ -greedy: greedy w.p. $1 - \epsilon + \epsilon/k$, each other ϵ/k .

UCB: $Q_t(a) + c\sqrt{\ln t / N_t(a)}$.

Gradient bandit: softmax on preferences $H(a)$.

MDPs & Returns

MDP: (S, A, P, R, γ) ; Markov: future depends only on current state.

Return: $G_t = \sum_{k \geq 0} \gamma^k R_{t+k+1}$; constant R : $G = R/(1 - \gamma)$.

Discount: $\gamma \in [0, 1)$ keeps sum finite; near 0 myopic, near 1 far-sighted.

Value Functions & Bellman

$V_\pi(s) = \mathbb{E}_\pi[G_t \mid S_t = s]$, $Q_\pi(s, a) = \mathbb{E}_\pi[G_t \mid s, a]$.

Bellman expectation: $V_\pi(s) = \sum_a \pi(a|s) \sum_{s'} p[r + \gamma V_\pi(s')]$.

Optimality: $V^*(s) = \max_a \sum_{s'} p[r + \gamma V^*(s')]$; $\pi^*(s) = \arg \max_a Q^*(s, a)$.

Dynamic Programming

Needs a model.

Policy iteration: evaluate $\rightarrow$ improve (greedy) until stable.

Value iteration: apply Bellman-optimality update until convergence, then act greedy.

GPI: evaluation + improvement interacting. Bellman operator is a γ -contraction (unique fixed point).

Monte Carlo & TD

MC: average actual returns over complete episodes; no model, no bootstrap; first-visit vs every-visit.

Control: estimate Q (no model); exploring starts or ϵ -soft; off-policy via importance sampling.

TD(0): $V(s) \leftarrow V(s) + \alpha[r + \gamma V(s') - V(s)]$ — bootstraps, online.

Exam tips & high-weightage: Discounted returns · Bellman equations · ϵ -greedy/UCB · value vs policy iteration · MC vs TD.